Health & Safety

Standard
Powered production equipment

Reference

Company	Domain	Document type	No. Sequence	Revision index	Langue
BYTP	H&S	REF	2233	B	EN

Revision	Date	Written by	Checked by	Approved by	Type of modification
A	24-07-2024	F.MARTIN	I.LELIEVRE O.BRZEZINSKI A.BERNARD	L.KNOLL	First issue
B	10-12-2025	I.MUCCIGNATO	F.MARTIN	L.KNOLL	Modifications following BYCN guidelines: Addiction of instructions regarding the setting up a work area and the minimum Requirements for workshop machines and the power tools

Table of contents

1. Objective

The purpose of this document is to outline general preventive and mitigation measures to ensure the safe use of powered production equipment on site.
This involves preventing the risks associated with these types of equipment in compliance with the regulations, codes and standards in force in all the countries where BYTP operates.

2. Scope

This document is focused on the mechanical risk of all powered machines and equipment that involves risk of cutting, perforating, crushing or entangling.

It does not cover the hazards and preventive measures linked to the energy sources driving the powered production tools, which are covered in the standard **Dangerous energies sources (BYTP-H&S-REF-2214)**.

This document applies to all companies working on sites (including service providers and subcontractors).
In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.
If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

3. Reference documents

- BYCN Guidelines - Powered production equipment
- BYTP-H&S-REF-2200 Standard Lines of defence and Life Saving Rules
- BYTP-H&S-REF-2010 Standard Lifting
- BYTP-H&S-REF-2012 Standard Vehicles pedestrian interaction
- BYTP-H&S-REF-2015 Standard Construction machinery and associated maintenance
- BYTP-H&S-REF-2018 Standard Ergonomics
- BYTP-H&S-REF-2020 Standard portable tools
- BYTP-H&S-REF-2023 Standard Confined spaces
- BYTP-H&S-REF-2024 Standard Work with high pressure tools
- BYTP-H&S-REF-2025 Standard Hot works
- BYTP-H&S-REF-2028 Standard Special works and deep foundations
- BYTP-H&S-REF-2030 Standard Work with Concrete
- BYTP and subsidiaries sensitive equipment policy
- Power tools official BYCN Guidelines
- Substitute guide for small angle grinder

4. Definitions and abbreviations

Powered production Equipment are Electrically, hydraulically, or pneumatically driven machinery and tools used on our worksites and workshops. These devices are designed to facilitate and expedite various construction tasks, such as cutting, drilling, lifting, or moving materials. Due to their operational mechanisms, they pose significant hazards including the risk of cutting, piercing, crushing or entangling workers, necessitating stringent safety protocols and protective measures.

A non-exhaustive list is provided here to illustrate the powered production equipment:

Power production tools

- Cutting Tools
  - Circular Saws
  - Chainsaws
  - Angle Grinders
  - Tile Cutters
  - etc.
- Drilling and Piercing Equipment
  - Drill Presses
  - Rotary Hammers
  - Handheld Power Drills
  - Nail Guns
  - etc.

Power production machinery

- Concrete Mixers
- Pumps
- Compactors and Rollers
- Vibratory Rollers
- Plate Compactors
- Generators and Compressors
- Portable Generators
- Air Compressors
- Welding and Cutting Machines
- etc.

This standard excludes:

- risks associated with the use of hand tools (hammers, crowbars, pliers, etc.), manual and electric screwdrivers, and drills
- risks associated with the use of site machinery used for lifting (loads or people), shovels, pallet trucks, etc. (excluding maintenance activities).

5. Roles and responsibilities

Each site implements the requirements of this document, in addition to any requirements that may be necessary to manage the risks specific to its environment.

Operational manager - The operational manager is an employee representing the employer and having the authority, the means and the competence to exercise decision-making power for the scope under its authority. He must enforce this Standard, monitor its implementation, ensure the proper equipment are provided to the workers and ensure that actions are properly taken as soon as possible when a deviation is identified.

Employees using powered production equipment - The Employees using powered production equipment are responsible for following safety protocols (including pre-use checks), using PPE, and reporting any defective equipment. They shall have the proper training, and appropriate certificate when required, to perform their activities using powered production equipment.

6. Training and competencies

Trainings

All personnel involved in the use of powered production equipment must be trained for its use. It includes the criteria for its visual inspection, the protocol in case of defect or damage, its handling in accordance with the manufacturer recommendation and on its storage.

All personnel involved in the installation operation, maintenance and removal of powered production machinery such as generators, pumps, conveyor must be trained and competent for the operation to be performed and certified when required.

Each operator may be subject to a skills assessment and effective practice of the machines, according to local requirements and regulations.

Competencies

Anyone assigned to an operation with a powered production equipment must be familiar with:

- Its handling and the related manufacturer operating procedures
- The criteria to verify that the equipment is in good condition or have the confirmation by a competent person
- The Safety devices that shall be in place or have the confirmation by a competent person
- The proper PPE to be worn
- The immediate Emergency response

7. Equipment and materials conformity

Equipment adequacy – Selection of each powered production equipment shall take into consideration its safety features and suitability for the intended tasks. Regular maintenance is equally important to prevent equipment failure and accidents. This includes routine inspections, timely repairs, and adherence to maintenance schedules.

Suitability and Condition of Consumables – Consumables must be adapted for each piece of equipment and maintained in good condition through appropriate storage in accordance with the manufacturer's technical documentation (selection, speed), and specifically using cutting discs with a thickness of 3,2mm

Provision of needs for consumables and specific PPE - For each powered production equipment, the required types of consumables and specific PPE must be identified and made available on site in sufficient quantities, with inventories subject to regular checks regarding their condition (expiry date, wear and tear, etc.).

Certificate of Compliance – Powered production equipment used on a project complies with local Regulations or in their absence, international regulations if any. This conformity is characterized by the provision of a certificate of conformity issued by the manufacturer himself, thus attesting to his commitment and his responsibility in respecting all the regulatory systems to which may be subject. the device he designed.

Machinery Periodic on-site inspections and General Periodic Inspection - All powered machinery used is uniquely identified, registered and subjected to a maintenance program defined considering local regulation, constructor’s recommendations, Client’s requirements and internal rules. All maintenance operations must be performed by a competent person or by an external competent company.

All point of concerns expressed during those inspections are lifted according to the instructions specified in the report. Those requiring immediate immobilization of the machinery are lifted before any (re)use.

All plants & equipment listed in the Policy regarding sensitive equipment BYTP and subsidiaries are subject to specific minimum requirements for frequency of periodic on-site inspections and General Periodic Inspection.
Verification before use - All Powered production equipment must be confirmed as safe to be used and in good condition for its commissioning and visually inspected before each use.
In case of damage, defects, or lack of identification (such as a number or equivalent), the powered production equipment must be put on standby. It should be removed from the worksite and segregated when possible. Additionally, a visual signal must be put in place to remind workers of the ban on its use.
Verification during visit and audit – Management of Powered production equipment is part of the topics that can be verified during on-site audits and visit onsite to ensure safety standards are upheld. These evaluations help identify areas for improvement and ensure compliance with regulations and industry standards.

8. Preparation of operations requiring the use of powered production equipment

8.1. Risk Assessment

Risk assessment of an operation include the identification of potential hazards associated with powered production equipment. This involves a thorough examination of the worksite, equipment, and tasks to capture specific risks.
Each captured risk must be evaluated to identify the routine and specific safety measures to be deployed according to its evaluation.
Each hazard must be documented and communicated to all relevant personnel involved in the activity, or exposed to it, during the pre-start briefing to ensure awareness and preparedness. The findings from the risk assessment should be systematically recorded and accessible.

It as to be noted that a continuous risks evaluating must be performed as conditions on the worksite can change rapidly. It is mandatory to stop an activity when an unidentified risk arise, or when the activity is not performed or cannot be performed as planned.

8.2. Setting up a work area

The setting up of work areas must be planned to ensure safety and ergonomics.
Spaces are designed to take into account the task environment and promote **stable working postures**. To protect operators against specific risks, safety devices are systematically deployed: this includes **screens** to protect against splashes and effective **collection systems for fumes or dust**, or any other safety device.
For **mobile workstations**, workbenches and vacuum cleaners suitable for each type of equipment must be provided. For workshops, clearly marked **exclusion zones** around the main equipment ensure that access is strictly reserved for authorized personnel.

8.3. Emergency response

Activities can only start once the Project emergency management and preparation procedure has been fully implemented.
This procedure must account for scenarios involving Powered production equipment and be developed in accordance with the recommendations of the **Standard Emergency Management (BYTP-H&S-REF-2221)**.
Compliance with these guidelines ensures that all potential risks are effectively managed before any operational activities begin.

9. Requirements for safe use of powered production equipment

At any time while using a powered production equipment, it is mandatory to apply the Life-Saving-Rules associated :

The use of powered production equipment on construction sites and workshops must be performed as per:

- The Regulation,
- The manufacturers / Constructors' recommendations
- Client requirement(s)
- Our internal requirements,

This ensures the highest levels of safety for all personnel.

9.1. Generic Requirements for production equipment

Internal requirements for a safe use of powered production equipment (e.g. circular saws, chainsaws, angle grinders, tile cutters) include:

- Equipment selection based on BYCN referenced equipment or, if not available, meets BYCN minimum Health and Safety requirements.
  Note: Use of grinders <230mm are prohibited (use substitution or derogation process)
- Operation covered in an operating procedure.
- Equipment was confirmed as adequate for the operation and certificate of compliance verified
- Operator receive appropriate and formalized training / on-site briefing prior using tools or equipment
- Operator is aware of the manufacturer recommendation on its use.
- Operation to be performed is subject to a risk assessment capturing the use of the powered equipment tool.
- Outcomes of the risk assessment is shared during the briefing of the operation
- Equipment is checked before each use and shall be in good condition.
- Safety devices, including accessible emergency stop devices, are in place and not shunted.
- Proper PPE are worn.
- When using powered production equipment on a workpiece, this last must be secured to prevent any dangerous movement.
- Operator and personnel around are aware of the immediate Emergency response.
- Regular audits and inspections are performed to ensure compliance with applicable requirements
- A program is in place to keep workers informed about the latest safety protocols.

Machinery such as concrete mixers, vibratory rollers and portable generators require special attention due to their size and power. In addition to previous safety measures, it must be ensured that :

- Equipment is signalized with appropriate labels which can include hazard signalization (e.g. crushing hazard, cutting hazard)
- The equipment must be secured to prevent its motion when it is not required for its installation, maintenance, use and removal from site.
- Exhaust, if any, such as gas, liquid or dust, are safely managed.

Specific and more detailed safety requirements and recommendations for Powered production equipment are presented on various Standards listed in section 3. Reference documents.

9.2. Minimum Requirements for workshop machines

- In order to ensure the safety of the operator**, all equipment must have its mobile parts inaccessible or protected by appropriate devices** (fixed, mobile or adjustable, light curtains, specific controls, etc.), and **all controls** (written in understandable language) must be **designed to prevent accidental maneuvering.**
- Equipment must **be started up voluntarily**, and each workstation must be equipped with **a safe emergency stop device.**
- To avoid **direct contact**, no live parts should be accessible, and any work on a machine should be carried out with the power off (machine disconnected).
- If the equipment or its use generates **hazardous materials or substances** (such as wood or silica dust, or welding fumes), appropriate capture devices must be put in place.
- Measures must be taken for electrical equipment to prevent any risk of burns related to **extreme temperatures.**
- **Safe access** to the machine is mandatory for all interventions (movements, adjustments, maintenance), including those that are necessary **during equipment operation**.

9.3. Minimum requirements for power tools

- The equipment must guarantee **compliance** with local regulatory requirements, as certified by the CE mark.
- The aim is to provide equipment that is as light as possible. The **weights** indicated correspond to the tool alone, without consumables (unless otherwise stated).
- The minimum **grip system** must ensure a good grip thanks to an ergonomic, non-slip handle (Soft-Grip), designed to make the tool comfortable to use without excessive strain or twisting of the wrist.
- The level of triaxial **vibration** specified by the supplier must be as close as possible to 2.5m/s² and comply with applicable standards, without ever exceeding 10 m/s² (limit for a maximum cumulative use of two hours per day).
- The minimum requirement **for protection against electric shock** is Class 2 (double insulation), except for core drilling tools, which must be Class 1 (requiring grounding).
- All power tools must be equipped with a dust **capture system at the source**, bearing in mind that certain groups of devices may require additional protection as identified in the BYCN tool catalog.
- **The noise exposure level** must not exceed 80dB(A) over an 8-hour period (including the effectiveness of PPE), with values specified by the supplier in accordance with applicable standards.
- The minimum **protection rating** requirement is IP20, with the exception of core drilling tools, which must have an IP55 rating.
- Depending on their type, power tools must incorporate specific **safety devices** such as a dead man's switch, a rapid braking system, and a protective housing (specified for each tool family).
- At the end of the shift, **the machine will be locked up** so that it does not remain in "self-service".
- Grinders smaller than 230 mm are prohibited; only the **FEIN 125 grinder** is authorized, and its replacement must be managed by consulting the **"Substitution Guide."**

- To ensure compliance with BYCN minimum requirements, **each country identifies its power tools by professional activity through a steering committee** that defines and updates a catalog of safe and ergonomic products. any use of a tool not included in the catalog requires **an exemption procedure** based on a comprehensive risk analysis and validated **by the health and safety manager of the operating unit**.